Wavepackets and multi (frequency) mode states

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Assumed Raouf has already done quantisation of fields if not it needs to come before this

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Disclaimer: lecture notes may contain "typo" type errors (factors of 2, minus signs, etc...) if something like this looks wrong there is a good chance it is, please let me know

1 Wavepackets of light

- Electric field operator for a single mode (G+K eq. 2.127):

$$\hat{\vec{E}}(z) = i\left(\frac{\hbar\omega}{2\epsilon_0 V}\right)^{\frac{1}{2}} \vec{e}_x[\hat{a}e^{i\vec{k}\cdot\vec{r}-i\omega t} - \hat{a}^\dagger e^{-i\vec{k}\cdot\vec{r}+i\omega t}] \tag{1}$$

- Only think about direction of propagation and a single polarisation, 1D problem

$$\hat{E}(z) = i\left(\frac{\hbar\omega}{2\epsilon_0 V}\right)^{\frac{1}{2}} [\hat{a}e^{ikz-i\omega t} - \hat{a}^\dagger e^{-ikz+i\omega t}] \tag{2}$$

- Infinite physical extent (in three dimensions, but let's focus on the direction of propagation)... but our devices have finite size, we need more modes!

- Multimode field in 1D

$$\hat{E}(z) = i\int_k dk \left(\frac{\hbar\omega}{2\epsilon_0 V}\right)^{\frac{1}{2}} [\hat{a}_k e^{ikz-i\omega_k t} - \hat{a}_k^\dagger e^{-ikz+i\omega_k t}] \tag{3}$$

- How do we build a wavepacket of finite extent $\rightarrow$ if we take $\langle E\rangle$ (and also note that in free space $\omega_k = ck$ where c is the speed of light)

$$\langle\hat{E}(z)\rangle = i\int_k dk \left(\frac{\hbar\omega}{2\epsilon_0 V}\right)^{\frac{1}{2}} [\langle\hat{a}_k\rangle e^{ikz-ickt} - \langle\hat{a}_k^\dagger\rangle e^{-ikz+ickt}] \tag{4}$$

- Propagates in the $+z$ direction at the speed of light c by definition, replace z with $z(t) = z - ct$

$$\langle \hat{E}(z) \rangle = i \int_k dk \left(\frac{\hbar\omega}{2\epsilon_0 V} \right)^{\frac{1}{2}} [\langle \hat{a}_k \rangle e^{ikz(t)} - \langle \hat{a}_k^\dagger \rangle e^{-ikz(t)}] \tag{5}$$

- Fun side-note/exercise, if we instead had $\omega_k = \frac{\hbar k^2}{2m}$, this would describe a (non-relativistic) massive particle, and speed would be $\frac{\partial \omega}{\partial k} = \frac{\hbar k}{m}$, the relationship between ω and k is known as the **dispersion relation** and is fundamental in almost all wave physics, light can also have more complicated dispersion in some materials

- Then this looks like a Fourier transform with $\langle \hat{a}_k \rangle$ and $\langle \hat{a}_k^\dagger \rangle$ as the Fourier coefficients $\rightarrow$ we can make wavepackets of arbitary shape; Problem? E is a physical observable, it must be real valued everywhere
  - $\langle \hat{a}_k \rangle$ and $\langle \hat{a}_k^\dagger \rangle$ are not independent!
  - Consider $|\psi\rangle = C_n|n\rangle + C_{n+1}|n+1\rangle$, $\langle \hat{a}_k \rangle_\psi = C_n C_{n+1}^\star$ but $\langle \hat{a}_k^\dagger \rangle_\psi = C_n^\star C_{n+1}$
  - Follows that $\langle \hat{a}_k^\dagger \rangle = \langle \hat{a}_k \rangle^\star$
  - Using the definition of sin and cosine[1] (and the fact that cosine is an even function):

$$\langle \hat{E}(z) \rangle = 2 \int_k dk \left(\frac{\hbar\omega}{2\epsilon_0 V} \right)^{\frac{1}{2}} [\mathrm{Re}(\langle \hat{a}_k \rangle) \sin(kz - ckt) - \mathrm{Im}(\langle \hat{a}_k \rangle) \cos(kz - ckt)] \tag{6}$$

  - Purely real by definition

2 Multimode wavepackets in practice

- A problem and its resolution:
  - Potential problem? Don't operators describing different frequency modes commute $[\hat{a}_k, \hat{a}_{k'}] = 0$ for $k \neq k'$? since we have a continuum of k how do we see any quantum effects at all?
  - Resolution: Instead of treating creation/annihilation operators for single modes, consider creation/annihilation for multimode wavepackets $[\hat{a}[\Psi], \hat{a}[\Phi]] \propto \langle \psi \mid \phi \rangle$, where $\hat{a}[\Psi] = \int dk \Psi(k) a_k$, overlap of wavefunctions is **distinguishability** $\rightarrow$ identical wavepackets (including phases, which indicate when they arrive) are indistinguishable

[1] In case you have forgotten, or not seen, these definitions: $\sin(x) = \frac{\exp(ix) - \exp(-ix)}{2i}$ and $\cos(x) = \frac{\exp(ix) + \exp(-ix)}{2}$

- Making indistinguishable wavepackets is key to many quantum optics applications, such as Boson sampling
- "Easy"[2] to measure indistinguishability technically with single photon wave packets, send both wavepackets into a beam splitter, measure how many times detectors click simultaneously, Hong-Ou-Mandel effect (see: https://arxiv.org/abs/quant-ph/0212017)

An uncertainty relationship between k and z:

- Taking the Fourier transform of k gives $k = -i\frac{\partial}{\partial z}$, taking $\langle \Delta k \rangle = \langle k^2 \rangle - \langle k \rangle^2 = -\int_z dz \frac{\partial^2 E^2(z)}{\partial z^2} + \int_z dz (\frac{\partial E^2(z)}{\partial z})^2$
  - n.b. I have used $\langle E^2(z) \rangle$ in my definition so it even works for states like number states where $\langle E(z) \rangle$ is zero everywhere.[3]
- Conceptually: a narrow peak requires a large negative curvature, therefore smaller Δz leads to larger Δk
- Mathematically: $[k, z] = -i\frac{\partial}{\partial z} z + iz\frac{\partial}{\partial z}$ by the product rule $-i\frac{\partial}{\partial z}(zE^2(z)) = -iE^2(z) - iz\frac{\partial}{\partial z}$, cancelling terms and taking expectation $\langle [k, z] \rangle = -i\int_z dz E^2(z)$
- Therefore $\langle \Delta k \Delta z \rangle \geq \int_z dz E^2(z)$ (note that $E^2(z)$ is not normalized like a wavefunction so can't just cancel to 1)
- Converse is not necessarily true, possible to have large (or even infinite) Δk and Δz simultaneously
- But wait! none of this depended on the system being quantum, isn't this just a statement about classical Fourier transforms which would also hold for very classical systems like water waves? Yes and no
  - The relationship between spread of position and momentum holds for classical systems
  - Can be thought of as a "classical uncertainty principle" but arguably not a "real" uncertainty principle since classical systems do not have projective measurements so all quantities can be known simultaneously
- Visible light has wavelength (λ) of the order 100s of nanometers $\sim 10^{-7} m$, plenty of space in experiments for many wavepackets even if $\Delta z \gg \lambda$
- Lots of other fun Fourier transform identities relating k and z I don't think they are relevant but we can discuss if you want

[2] relative to other experiments, none of this is really easy

[3] This has the complication that $\langle E^2(z) \rangle$ is non-zero for vacuum, but this can be mitigated by integrating over a finite extent and considering states of many photons so the vacuum contribution is small

3 Photons in wavepackets

- We can make a wavepacket with a single photon (a Gaussian wavepacket in this example)

$$|\psi_{\text{Gauss}}\rangle = \frac{1}{\mathcal{N}} \int_k dk \exp\left[-\frac{(k-k_0)^2}{\sigma^2}\right] \hat{a}_k^\dagger |0\rangle \tag{7}$$

- Superposition of frequencies, but all states involve only $(\hat{a}_k^\dagger)^1$ so only one photon
- Since an atom emits a photon over a finite period of time, it will always contain a superposition of frequencies (although Lorentzian/Voight, not Gaussian distributed)

$$|\psi_{\text{Lorentz}}\rangle = \frac{1}{\mathcal{N}} \int_k dk \frac{\gamma^2}{\gamma^2 + (k-k_0)^2} \hat{a}_k^\dagger |0\rangle \tag{8}$$

- Depending on the experiment may be justified in approximating as single frequency
- Can also have multi-photon distributions, for example a distribution of coherent states

$$|\psi_{\text{coh homo}}\rangle = \frac{1}{\mathcal{N}} \int_k dk \frac{\gamma^2}{\gamma^2 + (k-k_0)^2} \exp(\alpha \hat{a}_k^\dagger - \alpha^\star \hat{a}_k)|0\rangle \tag{9}$$

- The above state is a bit artificial, why? Once one photon is measured all others must have the same k (perfect correlation), this can happen (to a decent approximation) in a laser if all emitters are effected in the same way and is called **homogeneous broadening**
- If we assume photon frequencies are independent than we should have **inhomogeneous broadening**

$$|\psi_{\text{coh inhomo}}\rangle = \exp\left(\frac{1}{\mathcal{N}} \int_k dk \exp\left[-\frac{(k-k_0)^2}{\sigma^2}\right] (\alpha \hat{a}_k^\dagger - \alpha^\star \hat{a}_k)\right)|0\rangle \tag{10}$$

- Of course we can also have both

$$|\psi_{\text{broad}}\rangle = \frac{1}{\mathcal{N}'} \int_{k'} dk' \frac{\gamma^2}{\gamma^2 + (k'-k_0)^2} \exp\left(\int_k dk \exp\left[-\frac{(k-k')^2}{\sigma^2}\right] (\alpha \hat{a}_k^\dagger - \alpha^\star \hat{a}_k)\right)|0\rangle \tag{11}$$

- Which is a decent approximation of a wavepacket from a laser with Lorentzian homogenous broadening and Gaussian inhomogeneous broadening...
- If we wanted a continuous beam which is broadened rather than a wavepacket, we could add phases to the terms or consider an incoherent superposition of frequencies